BSRM X ALDO – RETAIL GEN AI HACKATHON

USE CASE:
RETHINKING RETURNS WITH AGENTIC AI

RETAIL GEN_AI HACKATH<>N

In partnership with ALDO GROUP

McGill | École Bensadoun de commerce au détail Bensadoun School of Retail Management

powered by aws

DATA SPHERE LAB

CONTEXT

Product returns are one of the most complex and costly parts of modern retail, particularly in footwear.

A return may look simple from the customer's perspective, but behind the scenes it can involve:

- **CUSTOMER EXPERIENCE AND INTERACTIONS**
- **E-COMMERCE**
- **STORES**
- **INVENTORY**
- **DISTRIBUTION CENTERS**
- **TRANSPORTATION**
- **MERCHANDISING**
- **PRODUCT QUALITY**
- **FINANCE**
- **FRAUD PREVENTION**
- **SUSTAINABILITY**

The reasons behind returns are equally varied. A product may not fit, may look different than expected, may arrive late or damaged, may simply no longer be wanted, or may be returned as a result of customer behaviors that negatively impact economics, including abuse or fraud.

Most retailers still manage returns through a combination of rules, workflows and disconnected systems. Decisions are often made without considering the full context of the customer, product, inventory position, logistics cost and likelihood of resale.

The business objective itself is simple: protect gross margin while improving customer satisfaction.

Agentic AI creates an opportunity to rethink this model as a whole.

An AI agent can potentially understand context, investigate information across multiple sources, make decisions, coordinate with people or systems, take actions and learn from outcomes.

THE CHALLENGE

How could agentic AI fundamentally improve the return lifecycle for both customers and retailers?

Design and prototype an agentic solution that addresses a meaningful problem related to retail returns across stores and e-commerce.

The objective is not simply to automate an existing returns workflow.

We encourage teams to rethink how returns could work when AI agents are able to understand context, reason about complex situations, make decisions and take action.

SCOPE

The scope of this challenge is intentionally broad. Returns should be considered as an end-to-end business problem rather than a single transaction or process. Teams are expected to determine where agentic AI can create the most meaningful value and define the problem they choose to solve.

Identifying the opportunity is part of the challenge.

WHAT MAKES THIS AN AGENTIC AI CHALLENGE?

Strong solutions should demonstrate some combination of:

- Understanding the end-to-end process, situation and objective
- Gathering information from multiple sources
- Reasoning about possible actions and trade-offs
- Making or recommending decisions
- Taking actions through APIs, tools or simulated business systems
- Coordinating multiple steps toward an objective
- Knowing when human approval is required
- Evaluating whether an action worked
- Adapting based on the result
- Demonstrating and defending a clear thesis for gross-margin protection and customer satisfaction

Teams may use one agent or multiple cooperating agents.

The use of generative AI alone does not make a solution agentic. We are looking for systems capable of reasoning and acting toward an objective, rather than simply generating content or answering questions.

YOUR SOLUTION

In the Ideation Phase 1: describe what your initial idea is for the development of a prototype to reduce returns (see Phase 1 Instrucions)

Finalists in the Final Hacking Phase: Teams should develop a working prototype demonstrating how their solution would operate.

Your prototype does not need to integrate with real ALDO systems.

Simulated APIs, databases and workflows may be used.

We care more about the quality of the problem identified, the strength of the team's thesis, the agent's reasoning and the potential business impact than production-ready software.

Each team should demonstrate the following 5 :

1. PROBLEM AND THESIS

Clearly explain:

- What problem or opportunity did you identify?
- Who or what is affected?
- Why does it matter?
- Why is the current approach insufficient?
- What is your thesis for improving gross margin and/or customer satisfaction?

2. AGENTIC APPROACH

Show:

- What the agent knows
- What it is trying to accomplish
- What decisions it makes
- What tools, data or systems it can access
- What actions it can take
- Where human intervention remains appropriate

3. WORKING PROTOTYPE

Demonstrate at least one end-to-end scenario.

The prototype should make the agent's reasoning, decisions and actions visible enough to understand how the solution works.

4. BUSINESS IMPACT

Quantify or estimate the potential impact of the solution.

Teams should select the measures most relevant to their own thesis rather than optimize against a predefined set of KPIs.

5. RESPONSIBLE AI

Explain the risks associated with giving the proposed agent autonomy and how those risks would be managed.

Consider issues such as:

- Customer fairness
- Privacy
- Bias
- Explainability
- Financial authority
- Appropriate human oversight

PHASE 1 JUDGING CRITERIA

BUSINESS IMPACT & STRENGTH OF THESIS – 30%

Has the team identified a meaningful problem or opportunity?

Is there a compelling and well-supported thesis for why the proposed solution creates value?

Is there a credible path to protecting gross margin and/or improving customer satisfaction?

AGENTIC INNOVATION – 25%

Does the solution genuinely take advantage of autonomous reasoning and action?

Would a traditional rules engine, analytics model or chatbot accomplish essentially the same thing?

SCALABILITY AND FEASIBILITY – 25%

Could this concept realistically operate at retail scale?

PRESENTATION – 20%

Is the solution clearly explained and convincingly demonstrated?

FINAL JUDGING CRITERIA

BUSINESS IMPACT & STRENGTH OF THESIS – 30%

Has the team identified a meaningful problem or opportunity?

Is there a compelling and well-supported thesis for why the proposed solution creates value?

Is there a credible path to protecting gross margin and/or improving customer satisfaction?

AGENTIC INNOVATION – 25%

Does the solution genuinely take advantage of autonomous reasoning and action?

Would a traditional rules engine; analytics model or chatbot accomplish essentially the same thing?

TECHNICAL EXECUTION – 15%

How effectively does the prototype demonstrate the concept?

SCALABILITY AND FEASIBILITY – 15%

Could this concept realistically operate at retail scale?

PRESENTATION – 15%

Is the solution clearly explained and convincingly demonstrated?

DATA AVAILABLE TO TEAMS

All datasets will be anonymized and may be simplified or synthetically enhanced for the hackathon.

Teams should not require access to production ALDO systems.

1. ORDER DATASET

Example fields:

- Order ID
- Customer ID
- Order date
- Channel
- Country
- Store or e-commerce
- SKU
- Product
- Colour
- Size
- Quantity
- Selling price
- Discount
- Promotion
- Shipping method
- Shipping cost
- Fulfilment location
- Delivery date
- Estimated delivery date

2. RETURNS DATASET

Example fields:

- Return ID
- Original order ID
- Customer ID
- SKU
- Return date
- Return channel
- Store or mail return
- Return reason
- Return sub-reason
- Refund amount
- Return shipping cost
- Product condition
- Disposition
- Days between purchase and return

Return reasons may include fields such as fit, comfort, quality, damage, expectations, delivery or change of mind.

3. CUSTOMER BEHAVIOR DATASET

An anonymized customer profile could include:

- Customer ID
- Country
- Loyalty status
- Customer tenure
- Number of lifetime orders
- Lifetime spend
- Number of returns
- Return rate
- Average order value
- Preferred channel
- Previous product categories purchased

No personally identifiable information would be provided.

DATA AVAILABLE TO TEAMS

4. PRODUCT CATALOGUE

Example fields:

- SKU
- Style
- Brand
- Category
- Subcategory
- Gender
- Colour
- Material
- Heel height
- Closure type
- Size
- Price
- Product description
- Product attributes
- Fit guidance
- Width
- Product type
- Heel type
- Sole material

5. PRODUCT REVIEWS

Example fields:

- Product
- Rating
- Review text
- Review date
- Size purchased
- Fit feedback
- Comfort rating
- Quality rating

Where available, structured fit information such as runs small, true to size or runs large may also be included.

6. INVENTORY DATASET

Inventory by location:

- SKU
- Location
- Location type
- On-hand quantity
- Available-to-promise quantity
- Incoming inventory

Locations may include distribution centers, stores and outlet stores.

7. STORE DATASET

Example fields:

- Store ID
- City
- Region
- Country
- Store format
- Inventory capacity
- Store operating characteristics

Precise customer addresses would not be provided.

8. LOGISTICS DATASET

Example fields:

- Origin
- Destination region
- Shipping method
- Estimated shipping cost
- Typical transit time
- Carbon estimate

This may be provided as a simplified transportation rate and service table.

DATA AVAILABLE TO TEAMS

9. PRODUCT ECONOMICS

Example fields:

- SKU
- Retail price
- Estimated product cost
- Gross margin
- Markdown level
- Estimated recovery value
- Processing cost

Commercially sensitive values may be normalized or synthetically generated while maintaining realistic economic relationships.

10. RETURN POLICIES AND BUSINESS RULES

A simplified set of policies covering areas such as:

- Return eligibility
- Final-sale rules
- Refunds
- Exchanges
- Loyalty-member exceptions
- Damaged merchandise
- Financial authority and approval limits

Some of this information may deliberately be provided as unstructured documents rather than structured data.

ADDITIONAL DATA

Depending on availability and anonymization requirements, teams may also receive:

- Product images
- Anonymized or synthetic customer-service conversations
- Store return-processing information
- External contextual datasets
- Other relevant retail data

Teams may also supplement the provided datasets with publicly available information where appropriate.

TECHNICAL ENVIRONMENT

Teams may be provided with simulated APIs representing retail capabilities such as retrieving information and executing transactions.

Examples could include capabilities to:

- Retrieve customer, order, product and inventory information
- Calculate transportation options
- Initiate or modify a return
- Create an exchange
- Issue a refund
- Route merchandise
- Modify selected product information

The environment is intended to allow agents to take actions, not simply generate recommendations.

Teams are not required to use every dataset or capability provided.